

ALIAH UNIVERSITY  
END TERM EXAMINATION, SPRING 2024  
PAPER: BCAUGMDC1201 (Engineering Mathematics I)  
All symbols have their usual meanings  
Use different answer scripts for different groups

Full Marks: 80

Time: 3 Hours

**GROUP - A (40 marks)**

Answer any **EIGHT** of the following questions:

$5 \times 8 = 40$

1. Show that the below sequence is convergent

$4 + 1$

$\sqrt{2}, \sqrt{2 + \sqrt{2}}, \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2}}}}, \dots$  Find out where it converges.

2. Define monotonic increasing sequence and monotonic decreasing sequence. Show

$1 + 1 + 3$

that the sequence  $\{u_n\}$  where  $u_n = \frac{5}{n+5}$  is monotonic increasing and bounded above.

3. Show that  $\lim_{n \rightarrow \infty} x^{\frac{1}{n}} = 1$ , for  $x > 0$ .

5

4. Define oscillatory series of Real numbers. Test the convergence of the series

$1 + 4$

$$\sum (n^3 + 1)^{\frac{1}{3}} - n.$$

5. Test the convergence of the series

5

$$1 + \frac{1}{2^2} + \frac{2^2}{3^3} + \frac{3^3}{4^4} + \dots$$

6. What is Global maximum and Global minimum? Find the value of  $x$  for which the function  $f(x) = x^3 - 9x^2 + 20x - 12$  has maximum and minimum. Find the global maximum of and minimum values in  $[0, 5]$ .

$1 + 4$

7. Write down the statement of D'Alembert's Ratio test. Test the convergence of the series

$1 + 4$

$$1 + \frac{3}{1!} + \frac{3^2}{2!} + \frac{3^3}{3!} + \frac{3^4}{4!} + \dots$$

8. Check the following series  $\sum u_n$  are convergent or divergent.

5

a)  $u_n = \frac{1}{\sqrt{n+5} - \sqrt{n}}$     b)  $u_n = \frac{n(n+5)}{(n+6)(n+7)}$ .

9. a) If  $f'(x)$  exists and  $< 0$  everywhere in  $(a, b)$ , then by using mean value theorem show that  $f(x)$  is a decreasing in  $(a, b)$

b) If  $f'(x)$  exists and  $> 0$  everywhere in  $(a, b)$ , then show that  $f(x)$  is increasing in  $(a, b)$ .

10. Show that the function  $f(x)$  is continuous at the points 5, 0, and -3, where,  $f(x) = |x| - |x+3| + |x-5|$ .

5

11. Evaluate by L' Hospital rule,  $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$ .

5

12. If  $m, n$  are positive integers, then

5

a) Show that  $B(m, n) = B(n, m)$ ,

b) Find the value of  $B(4, 2)$  using Gamma function.



GROUP - B (40 marks)

Answer any **EIGHT** of the following question;

$$5 \times 8 = 40$$

1. (a) If a line makes an angle  $\frac{\pi}{3}$  with  $x$ -axis,  $\frac{\pi}{3}$  with  $y$ -axis and  $\theta \in (0, \pi)$  with  $z$ -axis, then find the value of  $\theta$ . 3

- (b) Convert the point  $(7, -7\sqrt{3})$  to the polar coordinate. 2

2. Find the distance between the lines 5

$$\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-0}{0} \quad \text{and} \quad \frac{x-1}{0} = \frac{y-1}{1} = \frac{z-1.5}{1}$$

3. Find the equation of the plane passing through the point  $(1, 1, 1)$  and perpendicular to the planes  $x - y + z = 1$  and  $2x - 3y + z = 1$ . 5

4. Show that the four points  $(2, 3, -3)$ ,  $(1, 1, -2)$ ,  $(-1, 1, 4)$  and  $(1, 4, 1)$  lie on a plane. Also, write down the equation of the plane. 5

5. Find the nature of the quadric surfaces and hence evaluate its volume.

(a)  $2x^2 + 5y^2 + 3z^2 - 4x + 20y - 6z = 5$ . 3

(b)  $5x^2 + 5y^2 + 5z^2 - 10x - 10y + 10z = 45$ . 2

6. Find the curvature at any point  $t$  of the curve 5

$$F(t) = (t^2, \sin t, \cos t).$$

7. Find the torsion of the curve 5

$$F(t) = (t^2, 2t, t^3).$$

8. Evaluate

$$\iint dx dy$$

over the region bounded by a circle of radius 1, centre at  $(0, 0)$  and the line  $y = x$ . 5

9. Evaluate

$$\iint (x^2 + y^2) dx dy$$

over the region enclosed by the triangle having vertices  $(0, 0)$ ,  $(2, 0)$  and  $(1, 1)$ . 5

10. Evaluate

$$\iiint (x + y + z + 1) dx dy dz$$

over the region defined by  $x \geq 0$ ,  $y \geq 0$ ,  $z \geq 0$ ,  $x + y + z \leq 1$ . 5

11. Use Green's Theorem to evaluate

$$\oint_C (3xy^2 - 6y^2) dx + (y - 3xy) dy$$

where  $C$  is the boundary of the region  $x = 0$ ,  $y = 0$ ,  $2x + 3y = 6$ . 5

12. Use Divergence Theorem to evaluate

$$\iiint_S \vec{F} \cdot d\vec{S}$$

where  $\vec{F} = \sin(\pi x)\hat{i} + 2xy^3\hat{j} + (z^2 - 4x)\hat{k}$  and  $S$  is the surface of the box with  $-1 \leq x \leq 2$ ,  $0 \leq y \leq 1$  and  $1 \leq z \leq 4$ . 5